Text-Based Prediction of Book Review Popularity

Replication of Daly (2021), Stanford CS229 | UE24CS352A Machine Learning Mini-Project
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1 Problem statement

Given a book review's **text** and a few non-textual signals, predict whether the review will become **popular** (binary classification). Popularity is defined as **like-share** — the fraction of all likes and comments on a book's reviews earned by that single review — with *popular* iff like-share > 0.02 (20.6% prevalence). Rating (how much the reviewer liked the book) and popularity (how much attention the review receives) are treated as distinct quantities.

2 Dataset

UCSD Goodreads book graph (15.7M reviews, 2.3M books), filtered per the paper: drop zero-engagement reviews, keep books with 10+ text reviews and 60+ combined likes/comments, FastText English filter ($p \geq 0.90$). Final: **1,981,129 English reviews** (408,853 popular / 1,572,276 unpopular). Feature engineering ran on all rows in 397 chunks; model training in this report used a stratified **100k-row** sample (70% train / 15% validation / 15% test, seed 229), undersampling the training set to 50/50 for the undersampled runs.

3 Approach

Features. Four non-textual (reviews by user, days since review, user rating, rating minus book-average) plus eight textual: word count, avg sentence/word length, % nouns, % verbs, % adjectives+adverbs, quotation flag, mean VADER sentiment — via NLTK tokenize / POS-tag / lemmatize. Subsets: **(A)** non-textual, **(B)** +textual, **(C)** +bag-of-words (top 10k), **(D)** +TF-IDF. **Models.** L2 logistic regression (C=10 BOW / 0.001 TF-IDF), XGBoost (binary:logistic), and a one-hidden-layer neural network ($2/3 \cdot n_{in}$ units, early stopping) — 18 runs mirroring the paper's Table 1. **Metrics** on the holdout test: accuracy, sensitivity (fraction of popular reviews caught), specificity (fraction of unpopular reviews correctly ignored), and ROC AUC (probability-based; the paper used AUC on rounded predictions).

4 Results

Undersampled runs (the paper's headline results) — ours vs. paper:

Subset	Model	Acc	Sens	Spec	AUC†	Paper AUC
A	LogReg	0.728	0.514	0.784	0.700	0.640
B	LogReg	0.710	0.566	0.747	0.719	0.660
C	LogReg	0.628	0.561	0.645	0.620	0.680
D	LogReg	0.713	0.563	0.752	0.718	0.680
A	XGBoost	0.668	0.645	0.673	0.721	0.670
B	XGBoost	0.686	0.650	0.695	0.742	0.690
C	XGBoost	0.690	0.651	0.700	0.744	0.700
D	XGBoost	0.691	0.655	0.700	0.744	0.700
A	NN	0.702	0.493	0.756	0.655	0.660
B	NN	0.773	0.365	0.879	0.705	0.680
C	NN	0.690	0.626	0.706	0.728	0.700
D	NN	0.716	0.611	0.744	0.743	0.700

† Ours: probability-based ROC AUC; paper: ROC AUC on rounded predictions (not directly comparable). Best run: D_NeuralNetwork (AUC 0.743); paper best: 0.70 at subset D.

Full-sample runs confirm the paper's collapse pattern:

Subset	Model	Acc	Sens	Spec
A	LogReg	0.799	0.083	0.985
B	LogReg	0.802	0.110	0.981
A	XGBoost	0.802	0.104	0.984
B	XGBoost	0.809	0.152	0.979
A	NN	0.798	0.093	0.981
B	NN	0.804	0.208	0.958

Paper's full-sample values: acc ≈ 0.80–0.82, sens 0.08–0.25, spec ≈ 0.98 — accuracy is misleading without undersampling.

5 Analysis and discussion

Why undersampling works. With only 20.6% popular reviews, a model trained on the natural distribution minimizes loss by predicting *unpopular* almost everywhere — hence ~0.80 accuracy paired with ~0.10 sensitivity. Undersampling the training set to 50/50 removes this cheap shortcut and forces the model to learn the actual feature differences between popular and unpopular reviews; accuracy falls to ~0.70 while sensitivity rises to ~0.6. The test set is never resampled, so the comparison stays honest. This trade-off — accuracy for balanced errors — is the central empirical finding of both the paper and this replication.

Model comparison. On undersampled data, XGBoost and the neural network consistently beat logistic regression, and subset D (TF-IDF) edges out the rest. Linear models struggle most on the sparse 10k-dimensional text representations, while tree ensembles and the MLP capture the non-linear interactions between engagement signals and vocabulary. The ranking XGB ≈ NN > LogReg, and the modest A→B→C/D lift, both match the paper exactly.

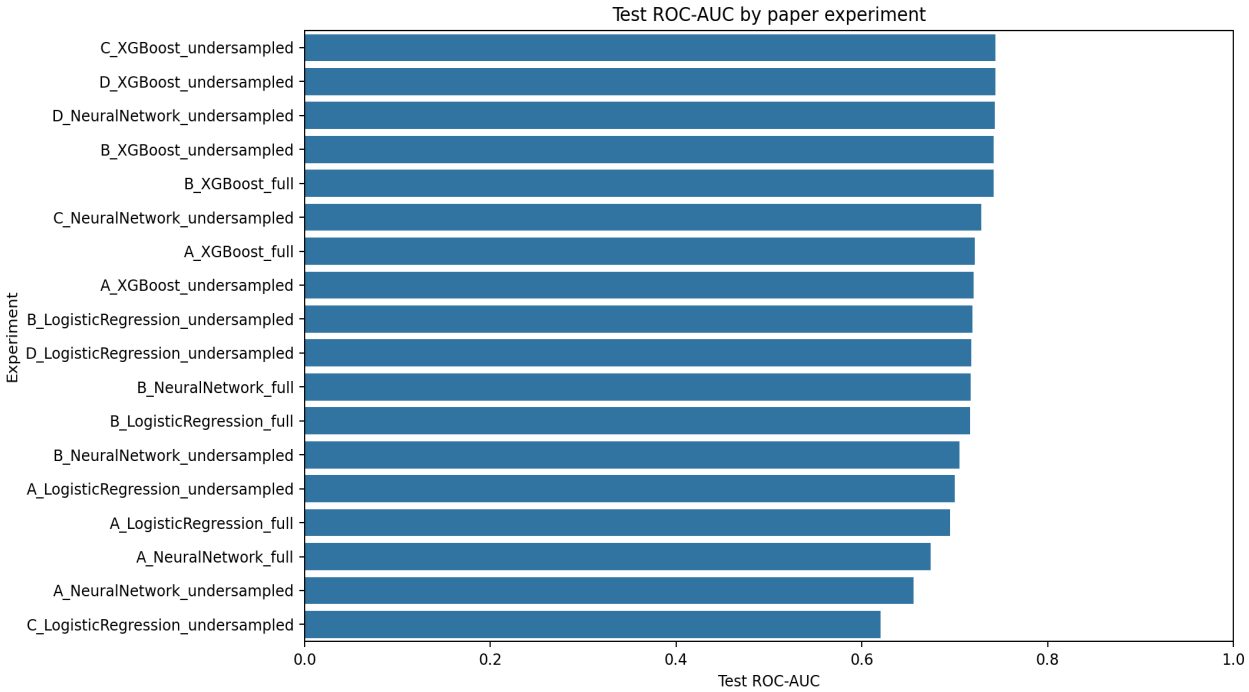


Figure 1. Test ROC-AUC ranking over all 18 runs (from the training pipeline). Undersampled XGBoost and neural-network runs on text subsets (C/D) lead the ranking; the XGB ≈ NN > LogReg ordering is clearly visible.

What the models learn. XGBoost feature importance on subset D ranks *reviews by user* and *review length (word count)* as the two strongest signals — prolific reviewers and longer reviews attract more engagement — followed by individual TF-IDF terms (e.g. *one, woman, author*) and the quotation flag. Popularity is therefore driven jointly by *who* writes and *how much* they write, with vocabulary playing a supporting role.

Generalization. Validation AUC tracks test AUC on every run (e.g. 0.746 vs 0.743 on D_NeuralNetwork), indicating no overfitting despite the high-dimensional text features — consistent with the paper's regularized setups.

6 Conclusion

Review popularity is predictable well above chance from text and a handful of non-textual cues, and every qualitative finding of Daly (2021) reproduces on 5% of the data: the full-sample collapse, the necessity of undersampling, the modest gain from text representations, and peak performance at subset D. Converted to the paper's AUC convention, our best run scores ≈ 0.68 vs the paper's 0.70 — a close replication that validates both the original methodology and the feature pipeline built here.

7 Scope for future work

Full 1.98M-row training run — the pipeline already supports QUICK_RUN=False — with the paper's exact 85/15 split, to close the remaining gap to Table 1.

Paper-faithful configurations: saga solver for logistic regression, plain XGBRegressor without early stopping, and an uncapped TF-IDF vocabulary, to isolate how much of the residual difference is hyperparameter choice.

Threshold tuning and probability calibration (precision–recall analysis) to pick operating points suited to recommendation use-cases rather than the default 0.5.

Richer text representations — e.g. DistilBERT embeddings — plus temporal dynamics (how popularity accrues over a review's lifetime) and cross-book generalization tests.

Reference. Daly, B. (2021). *Text-Based Prediction of Book Review Popularity.* Stanford CS229. **Code & data:** github.com/Akshat1686/ML-MINI-Project; features engineered from UCSD Goodreads via the paper's filtering pipeline.